23CSE202 DATABASE MANAGEMENT SYSTEMS
S3 B.Tech CSE
Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus

PROJECT ABSTRACT

Group No : A5

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Project Title :

Hotel Room Booking and Hospitality Management System

Abstract:

Hotels, resorts, and guest houses continue to face operational challenges due to reliance on manual registers or disconnected software tools for managing bookings, guest records, billing, and staff coordination. This often results in double bookings, inaccurate room availability, delayed check-in/check-out, billing errors, and poor visibility into occupancy and revenue trends, making it difficult for management to make timely, data-driven decisions. This project proposes a centralized Hotel Room Booking and Hospitality Management Database System, built on a relational database, to unify all core hotel operations under a single, normalized structure that ensures accuracy, consistency, and quick retrieval of information. The primary objective is to automate customer registration, room inventory management, reservation handling, check-in/check-out processing, billing, and staff record-keeping. Customers can browse and reserve rooms based on category, availability, and pricing, while receptionists manage bookings and guest interactions in real time. Administrators oversee occupancy, revenue, and staff, managers generate performance reports, and housekeeping staff update room status after cleaning or maintenance, keeping availability data accurate at all times. Intended users include hotel administrators, receptionists, managers, housekeeping staff, and customers, each operating under defined roles and access levels. The database consists of interrelated entities - Customer, Room, Reservation, Employee, Department, Payment, Invoice, and Room Service - linked through primary and foreign key relationships to maintain referential integrity and eliminate redundancy through normalization. The project incorporates database complexity beyond basic CRUD operations. Multi-table joins combine Customer, Room, Reservation, Payment, and Room Service data to produce a guest's complete stay summary in one query. Room availability is validated using date-range overlap logic rather than a simple status check, ensuring no room is double-booked. Triggers automatically update room status, marking rooms "occupied" on confirmed booking and "needs cleaning" on check-out, removing manual updates. A stored procedure handles the full check-out process, including bill calculation, invoice generation, payment update, and room reset, as a single atomic operation, with transaction handling ensuring reservation and payment updates commit or roll back together.

Aggregate queries support occupancy-rate analysis and monthly revenue summaries, while a view exposes real-time room availability computed dynamically rather than stored redundantly. CHECK and UNIQUE constraints enforce data validity, such as preventing invalid date ranges and overlapping bookings for the same room. By integrating these features, the system aims to reduce manual errors, improve operational efficiency, enhance the guest experience, and provide reliable, real-time data for decision-making. The scope and complexity of this project, spanning normalization, triggers, stored procedures, transactions, and reporting, make it well-suited for completion within a two-month timeline, offering a practical and feasible submission for the Database Management Systems course project.